SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA1610 COURSE NAME: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)

QUESTIONS: 5 **Total Marks:50**

Annexure A

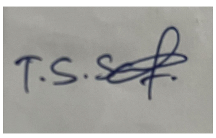
Experiments

Criteria	Understanding of Concepts ***(2.5)***	Experimental Design ***(3)***	Use of Tools ***(2)***	Analysis of Results ***(1.5)***	Documentation ***(1)***	***Total(10)***
Weightage	*25%*	30%	20%	15%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I T.S.Sabarish 192371014 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Annexure C

Experiments :

Children of three ages are asked to indicate their preference for three photographs of adults. Do the data suggest that there is a significant relationship between age and photograph preference? What is wrong with this study? (10 Marks)

	Photograph:		
Age of child	A	B	C
5-6 years:	18	22	20
7-8 years:	2	28	40
9-10 years:	20	10	40

Use cov() to calculate the sample covariance between B and C.

Use another call to cov() to calculate the sample covariance matrix for the preferences.

Use cor() to calculate the sample correlation between B and C.

Use another call to cor() to calculate the sample correlation matrix for the preferences.

2. Assume the Tennis coach wants to determine if any of his team players are scoring outliers. To visualize the distribution of points scored by his players, then how can he decide to develop the box plot? (10 Marks)

Player ID	**Player Name**	**Points Scored**
P1	Player A	12
P2	Player B	15
P3	Player C	14
P4	Player D	10
P5	Player E	18
P6	Player F	20
P7	Player G	22
P8	Player H	13
P9	Player I	11
P10	Player J	35
P11	Player K	16
P12	Player L	17
P13	Player M	19
P14	Player N	21
P15	Player O	23

3. A bank wants to decide whether to approve a loan based on customer attributes like:

- Age
- Income
- Employment Status
- Credit Score

Using historical data, build a **Decision Tree classifier** to predict **Loan Approval (Yes/No)**.
(10 Marks)

Consider the Data Set
young,high,employed,good,yes
young,high,self-employed,average,yes
middle,high,employed,good,yes
old,medium,employed,good,yes
old,low,unemployed,poor,no
old,low,self-employed,average,no
middle,low,unemployed,poor,no
young,medium,employed,average,yes
young,low,unemployed,poor,no
old,medium,self-employed,good,yes
middle,medium,employed,average,yes
middle,high,self-employed,good,yes
old,medium,unemployed,poor,no
young,medium,self-employed,average,yes
middle,low,employed,poor,no

4. Two Maths teachers are comparing how their Year 9 classes performed in the end of year exams. Their results are as follows: (5 Marks)
Class A: 76, 35, 47, 64, 95, 66, 89, 36, 8476,35,47,64,95,66,89,36,84
Class B: 51, 56, 84, 60, 59, 70, 63, 66, 5051,56,84,60,59,70,63,66,50
(i) Find which class had scored higher mean, median and range.
(ii) Plot above in boxplot and give the inferences

5. Suppose that a hospital tested the age and body fat data for 18 randomly selected adults with the following results: (10 Marks)

age	23	23	27	27	39	41	47	49	50
%fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2

age	52	54	54	56	57	58	58	60	61
%fat	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

(a) Calculate the mean, median, and standard deviation of *age* and *%fat*.
(b) Draw the boxplots for *age* and *%fat*.
(c) Draw a *scatter plot* and a *q-q plot* based on these two variables.

Question 1 — Age vs Photograph Preference (10 Marks)

Problem: Children of three age groups indicated their preference for three photographs (A, B, C) of adults. We must determine whether age and photograph preference are significantly related, and identify flaws in the study design.

Observed Data

Age of Child	Photograph A	Photograph B	Photograph C	Row Total
5–6 years	18	22	20	60
7–8 years	2	28	40	70
9–10 years	20	10	40	70
Column Total	40	60	100	200

Test of Significant Relationship — Chi-Square Test of Independence

Since both variables (Age group and Photograph preference) are categorical, the appropriate test for association is the Chi-Square (χ^2) test of independence.

H_0 (Null hypothesis): Age and photograph preference are independent (no relationship).

H_1 (Alternative hypothesis): Age and photograph preference are NOT independent (a relationship exists).

Expected frequency for each cell = (Row Total × Column Total) / Grand Total

Age of Child	Expected A	Expected B	Expected C
5–6 years	12.00	18.00	30.00
7–8 years	14.00	21.00	35.00
9–10 years	14.00	21.00	35.00

$\chi^2 = \Sigma$ [(Observed − Expected)² / Expected] = 29.60

- Degrees of freedom, df = (rows − 1)(cols − 1) = (3−1)(3−1) = 4
- Critical value of χ^2 at 5% significance level, df = 4 → 9.488
- Calculated χ^2 (29.60) > Critical χ^2 (9.488) ⟹ Reject H_0

Conclusion: There IS a statistically significant relationship between the age of the child and photograph preference ($p < 0.001$). Younger children (5–6) show a more even spread, the 7–8 group strongly favours photographs B and C over A, and the 9–10 group favours C most.

Covariance and Correlation using R

Treating each photograph's preference counts across the 3 age groups as a variable (A, B, C), we can build a preference matrix and use R's cov() and cor() functions:

```
# Preference data (rows = age groups, columns = photographs)
A <- c(18, 2, 20)
B <- c(22, 28, 10)
C <- c(20, 40, 40)
pref <- cbind(A, B, C)

# (i) Sample covariance between B and C
cov(B, C)

# (ii) Sample covariance matrix for A, B, C
cov(pref)

# (iii) Sample correlation between B and C
cor(B, C)

# (iv) Sample correlation matrix for A, B, C
cor(pref)
```

Results (computed):

Quantity	Value
cov(B, C)	-20.00
cor(B, C)	-0.189

Covariance matrix cov(A,B,C):

	A	B	C
A	97.33	-74.00	-46.67
B	-74.00	84.00	-20.00
C	-46.67	-20.00	133.33

Correlation matrix cor(A,B,C):

	A	B	C
A	1.000	-0.818	-0.410
B	-0.818	1.000	-0.189
C	-0.410	-0.189	1.000

Interpretation: A and B are strongly negatively correlated (-0.818) — as preference for A rises across groups, preference for B falls. B and C are weakly negatively correlated (-0.189). A and C are moderately negatively correlated (-0.410).

What is Wrong with this Study?

- Very small / uneven cell counts — the cell '7–8 years, Photo A = 2' is extremely small, making the chi-square approximation unreliable for that cell (expected count rule of thumb: expected ≥ 5 in every cell is preferred).
- No random sampling / random assignment described — children may not represent the general population of each age group.

- Confounding variables not controlled — gender, cultural background, and familiarity with the photographed adults could influence preference.
- Age grouping is arbitrary — collapsing continuous age into 3 bins can hide or exaggerate real trends (ecological/aggregation bias).
- Only 3 photographs used — limited stimuli reduce generalizability of 'photograph preference' as a construct.
- No repeated trials / no measure of preference strength — a single forced choice does not capture intensity of preference.
- Chi-square shows association, not causation — the test cannot explain WHY age relates to preference.

Question 2 — Boxplot for Outlier Detection in Tennis Scores (10 Marks)

Problem: The coach wants to detect scoring outliers among 15 players and visualise the distribution of points scored using a boxplot.

Data

Player ID	P1	P2	P3	P4	P5	P6	P7	P8
Points	12	15	14	10	18	20	22	13

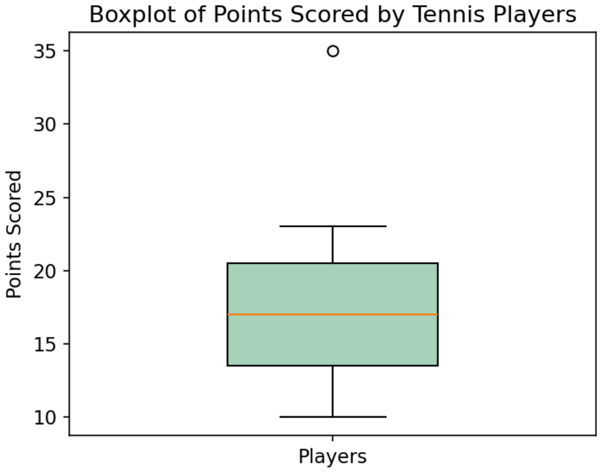
Player ID	P9	P10	P11	P12	P13	P14	P15
Points	11	35	16	17	19	21	23

Steps to Construct the Boxplot

- Step 1 — Sort the data: 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 35
- Step 2 — Find the median (Q2): middle value = 17
- Step 3 — Find Q1 (median of lower half) = 13.5 and Q3 (median of upper half) = 20.5
- Step 4 — Compute IQR = Q3 − Q1 = 20.5 − 13.5 = 7.0
- Step 5 — Compute fences: Lower fence = Q1 − 1.5×IQR = 13.5 − 10.5 = 3.0; Upper fence = Q3 + 1.5×IQR = 20.5 + 10.5 = 31.0
- Step 6 — Any point outside [3.0, 31.0] is plotted as an outlier. Here, Player J = 35 exceeds the upper fence ⟹ Player J is a scoring outlier.

R code to build the boxplot:

```
points <- c(12,15,14,10,18,20,22,13,11,35,16,17,19,21,23)
boxplot(points,
        main = "Boxplot of Points Scored by Tennis Players",
        ylab = "Points Scored", col = "lightgreen")
boxplot.stats(points)$out   # returns outlier values -> 35
```



Decision

Q1 = 13.5, Median = 17, Q3 = 20.5, IQR = 7.0, Mean ≈ 17.73, Std. Dev ≈ 6.25.

The coach can conclude that Player J's score of 35 lies well beyond the upper fence (31.0) and stands out as a statistical outlier — either an exceptionally strong performance or a data-entry anomaly worth verifying. All other players fall within the expected range of variation.

Question 3 — Decision Tree Classifier for Loan Approval (10 Marks)

Problem: Build a Decision Tree classifier (ID3 algorithm, using Information Gain) to predict Loan Approval (Yes/No) from Age, Income, Employment Status and Credit Score.

Training Data

#	Age	Income	Employment	Credit Score	Loan Approved
1	young	high	employed	good	yes
2	young	high	self-employed	average	yes
3	middle	high	employed	good	yes
4	old	medium	employed	good	yes
5	old	low	unemployed	poor	no
6	old	low	self-employed	average	no
7	middle	low	unemployed	poor	no
8	young	medium	employed	average	yes
9	young	low	unemployed	poor	no
10	old	medium	self-employed	good	yes
11	middle	medium	employed	average	yes
12	middle	high	self-employed	good	yes
13	old	medium	unemployed	poor	no
14	young	medium	self-employed	average	yes
15	middle	low	employed	poor	no

Class distribution: 9 records = "yes", 6 records = "no" (n = 15)

Step 1 — Entropy of the Whole Dataset

Entropy(S) $= -\Sigma\, p_i \log_2(p_i)$

Entropy(S) $= -(9/15)\cdot\log_2(9/15) - (6/15)\cdot\log_2(6/15) = 0.4421 + 0.5288 = 0.971$ **bits**

Step 2 — Information Gain for Each Attribute

Information Gain(S, Attribute) = Entropy(S) $- \Sigma\, (|S_v|/|S|) \times$ *Entropy*(S_v) *for each value v of the attribute*

Attribute	Weighted Remaining Entropy	Information Gain
Age	0.888	0.083
Income	0.260	0.711

Attribute	Weighted Remaining Entropy	Information Gain
Employment Status	0.501	0.470
Credit Score	0.241	0.730

Credit Score gives the highest Information Gain (0.730) ⟹ it becomes the ROOT node.

Step 3 — Split on Credit Score

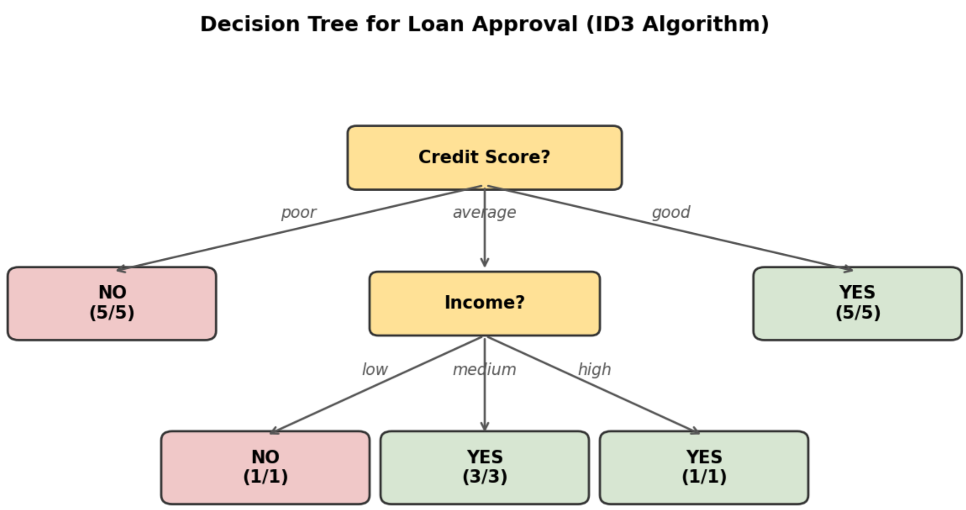
- Credit Score = poor → {5 records, all "no"} ⟹ Pure node ⟹ Leaf = NO
- Credit Score = good → {5 records, all "yes"} ⟹ Pure node ⟹ Leaf = YES
- Credit Score = average → {5 records: 4 yes, 1 no} ⟹ Not pure, needs further splitting. Entropy = 0.722 bits

Step 4 — Split the "Average" Credit-Score Branch

Within the Credit Score = average subset, Information Gain is recalculated for the remaining attributes (Age, Income, Employment). Income (and Age, which ties) gives the highest gain; splitting on Income cleanly separates the classes:

- Income = low → {1 record: no} ⟹ Leaf = NO
- Income = medium → {3 records: yes, yes, yes} ⟹ Leaf = YES
- Income = high → {1 record: yes} ⟹ Leaf = YES

Final Decision Tree



Decision Rules extracted from the tree:

- IF Credit Score = good → Loan Approved = YES
- IF Credit Score = poor → Loan Approved = NO
- IF Credit Score = average AND Income = low → Loan Approved = NO
- IF Credit Score = average AND Income ∈ {medium, high} → Loan Approved = YES

R code (using the rpart / party package):

```
library(rpart)
library(rpart.plot)
loan <- data.frame(
  Age = c("young","young","middle","old","old","old","middle","young",
          "young","old","middle","middle","old","young","middle"),
  Income = c("high","high","high","medium","low","low","low","medium",
             "low","medium","medium","high","medium","medium","low"),
  Employment = c("employed","self-employed","employed","employed","unemployed",

"self-employed","unemployed","employed","unemployed","self-employed",

"employed","self-employed","unemployed","self-employed","employed"),
  CreditScore =
c("good","average","good","good","poor","average","poor","average",
                  "poor","good","average","good","poor","average","poor"),
  Approved = c("yes","yes","yes","yes","no","no","no","yes",
               "no","yes","yes","yes","no","yes","no")
)
model <- rpart(Approved ~ Age + Income + Employment + CreditScore,
                data = loan, method = "class")
rpart.plot(model)
```

Question 4 — Comparing Class A and Class B Exam Results (5 Marks)

Problem: Two Year-9 Maths classes are compared on end-of-year exam scores.

Data

Class A
76, 35, 47, 64, 95, 66, 89, 36, 84

Class B
51, 56, 84, 60, 59, 70, 63, 66, 50

(i) Mean, Median and Range

Statistic	Class A	Class B	Higher
Mean	65.78	62.11	Class A
Median	66.00	60.00	Class A
Range (Max − Min)	95 − 35 = 60	84 − 50 = 34	Class A

Class A scored higher on all three measures — mean, median, and range. Class A's much larger range (60 vs 34) also shows its scores are far more spread out / less consistent than Class B's.

(ii) Boxplot and Inference

Statistic	Class A	Class B
Minimum	35	50
Q1	47	56
Median (Q2)	66	60
Q3	84	66
Maximum	95	84
IQR	37	10

```
classA <- c(76, 35, 47, 64, 95, 66, 89, 36, 84)
classB <- c(51, 56, 84, 60, 59, 70, 63, 66, 50)
boxplot(classA, classB, names = c("Class A","Class B"),
        col = c("orange","skyblue"),
        main = "Boxplot Comparison: Class A vs Class B",
        ylab = "Exam Score")
```

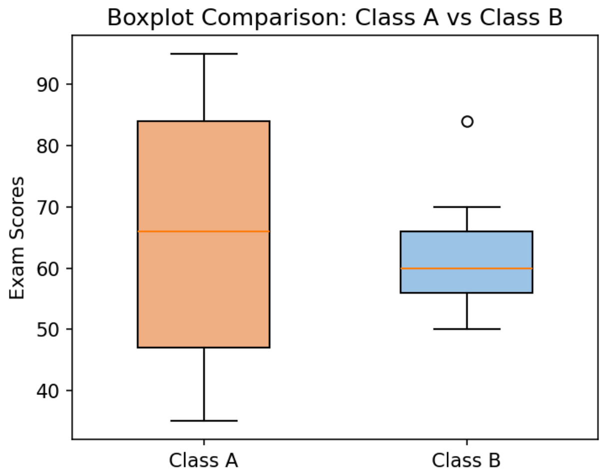


Fig 4.1: Boxplot comparison of Class A and Class B exam scores.

Inferences:

- Class A has a higher median score and a wider spread (IQR = 37 vs 10) — its box is taller, indicating greater variability in student performance.
- Class B's scores are more tightly clustered around the median (IQR = 10), suggesting more consistent performance across students.
- Class A's box is skewed — the median (66) is closer to Q3 (84) than Q1 (47), suggesting a left (negative) skew, i.e. more students scored above the median with a few low scorers pulling the range down.
- No outliers appear beyond the whiskers for either class (within 1.5×IQR of the quartiles), so all scores are considered typical for their class.
- Overall: Class A performed better on average but less uniformly; Class B performed slightly lower on average but more uniformly.

Question 5 — Age and Body Fat (%fat) Analysis (10 Marks)

Problem: A hospital recorded age and %body-fat for 18 randomly selected adults.

Data

age	23	23	27	27	39	41	47	49	50
%fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2

age	52	54	54	56	57	58	58	60	61
%fat	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

(a) Mean, Median, Standard Deviation

Statistic	Age	%fat
Mean	46.44	28.78
Median	51.00	30.70
Standard Deviation (sample)	13.22	9.25

```
age <- c(23,23,27,27,39,41,47,49,50,52,54,54,56,57,58,58,60,61)
fat <-
c(9.5,26.5,7.8,17.8,31.4,25.9,27.4,27.2,31.2,34.6,42.5,28.8,33.4,30.2,34.1,32.9,41
.2,35.7)
mean(age); median(age); sd(age)
mean(fat); median(fat); sd(fat)
```

(b) Boxplots for Age and %fat

```
boxplot(age, main = "Boxplot of Age", ylab = "Age (years)", col = "gold")
boxplot(fat, main = "Boxplot of %fat", ylab = "Body Fat (%)", col = "orchid")
```

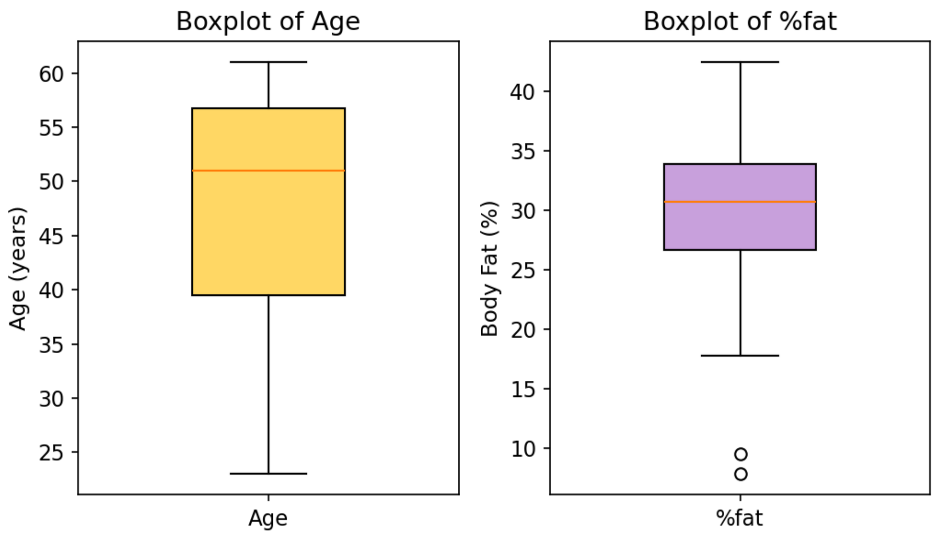


Fig 5.1: Boxplots of Age (left) and %fat (right). The two low %fat values (7.8, 9.5) appear as mild outliers below the lower whisker.

(c) Scatter Plot and Q-Q Plot

```
plot(age, fat, main = "Scatter Plot: Age vs %fat",
     xlab = "Age", ylab = "%fat", pch = 19, col = "blue")
abline(lm(fat ~ age), col = "red", lty = 2)
cor(age, fat)

qqplot(age, fat, main = "Q-Q Plot: Age vs %fat",
       xlab = "Age quantiles", ylab = "%fat quantiles")
```

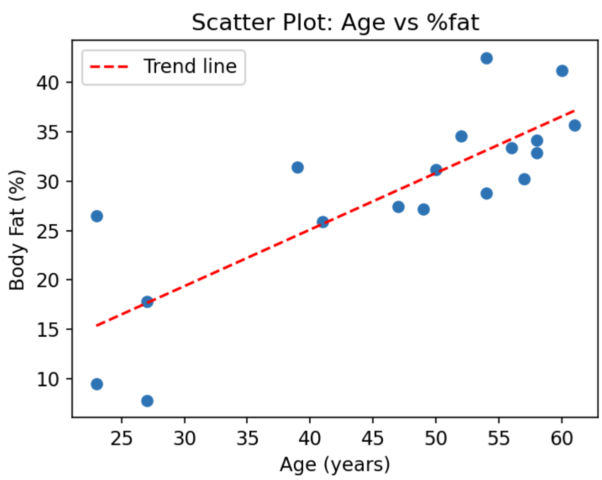


Fig 5.2: Scatter plot of Age vs %fat with fitted trend line (r ≈ 0.818, strong positive linear relationship).

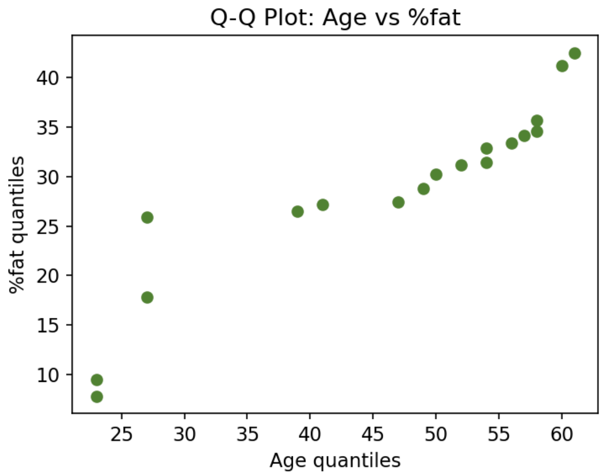


Fig 5.3: Q-Q plot comparing the quantiles of Age and %fat — the near-linear pattern confirms the two distributions have a similar shape/relationship.

Interpretation:

- The scatter plot shows a strong positive linear relationship between age and body fat (r ≈ 0.82) — %fat tends to rise as age increases.
- The boxplot of %fat shows two mild low outliers (7.8% and 9.5%), corresponding to the two youngest adults (age 23, 27) with unusually low body fat.
- The Q-Q plot's roughly straight/linear pattern indicates that the quantiles of age and %fat increase together at a fairly constant rate, reinforcing the strong positive association seen in the scatter plot.
- Age is fairly symmetric (mean 46.44 ≈ median 51 loosely), while %fat is close to symmetric with a median (30.7) slightly above the mean (28.78), suggesting a mild left skew caused by the two low outlier values.